

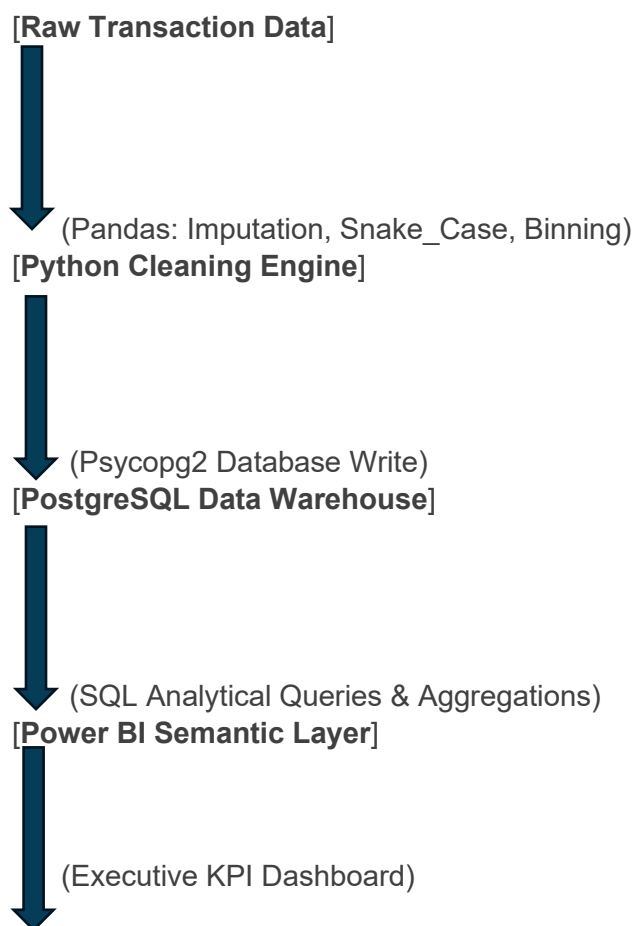
CUSTOMER SHOPPING BEHAVIOR ANALYSIS

June 16 2026

PROJECT OVERVIEW

This report transforms the transactional data of 3,900 unique purchases into actionable strategic intelligence. By integrating end-to-end data processing—from programmatic Python cleaning to PostgreSQL relational warehousing and Power BI visual semantic layers—this analysis isolates the friction points and revenue engines dictating customer lifetime value.

TECHNICAL PIPELINE ARCHITECTURE



1. CORE PLATFORM PERFORMANCE METRICS

An evaluation of the platform's baseline operational metrics reveals consistent transactional value across customer segments, despite structural imbalances in user acquisition.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	3900
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	3900
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	22
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN

Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900	3900.000000	3900	3900
2	2	NaN	6	7
No	No	NaN	PayPal	Every 3 Months
2223	2223	NaN	677	584
NaN	NaN	25.351538	NaN	NaN
NaN	NaN	14.447125	NaN	NaN
NaN	NaN	1.000000	NaN	NaN
NaN	NaN	13.000000	NaN	NaN
NaN	NaN	25.000000	NaN	NaN
NaN	NaN	38.000000	NaN	NaN
NaN	NaN	50.000000	NaN	NaN

2. DEEP-DIVE SQL BEHAVIORAL INSIGHTS

Demographic Revenue Skew

An extraction of demographic data fields exposes a significant imbalance in purchasing power across gender definitions:

	gender text	revenue numeric
1	Female	75191
2	Male	157890

Strategic Realignment: The business is highly dependent on male consumer segments. Marketing portfolios must execute a dual strategy: protect the core male purchasing engine while deploying targeted acquisition campaigns to capture market share in under-indexed female verticals.

The Subscription Paradox:

Cross-referencing membership status against direct user spend values highlights a structural anomaly in user behavior:

Subscription Status Revenue Contribution Breakdown:

[Subscribers: 1,053 Users] —► Gross Revenue: \$62,645.00 (Avg Spend: \$59.49)

[Non-Subscribers: 2,847 Users] —► Gross Revenue: \$170,436.00 (Avg Spend: \$59.87)

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

While non-subscribers generate the clear majority of gross transactional volume (\$170,436), their average ticket size (\$59.87) matches standard subscriber spending (\$59.49) almost exactly. Members are not buying higher-value items or paying premium ticket margins.

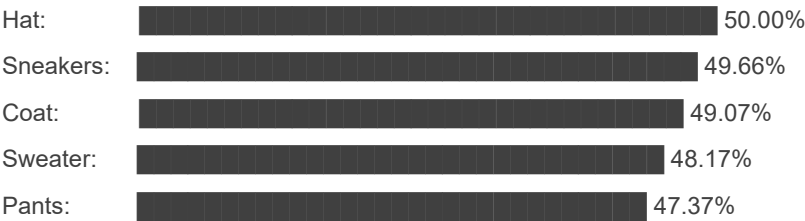
The primary business value of the subscription program is **predictable transaction frequency**, rather than expanded cart sizes.

Value-Driven Loyalty Vulnerability

Filtering for transactions that exceed the platform's average spend threshold (\$59.76) while simultaneously utilizing promotional codes yields **839 high-value, discount-dependent customers**.

At the product tier, specific inventory lines show a structural reliance on promotions to drive sales volume:

Discount Penetration Rate by Core Product Lines:



	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

This high promotional volume poses a risk to profit margins. Nearly half of all sales for top items like hats and sneakers only occur because a discount is attached. The platform must restructure its pricing strategy to decouple these anchor products from constant promotional support.

3. PRODUCT & INVENTORY PERFORMANCE MATRIX

Analyzing product rankings by order volume across major categories reveals clear consumer preferences. This matrix highlights which specific items act as primary entry points for each department:

Core Category Product Performance

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessories	Jewelry	171
2	2	Accessories	Sunglasses	161
3	3	Accessories	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

4. OPERATIONAL & DELIVERY LOGISTICS ANALYSIS

Logistical settings reveal a direct connection between fulfillment speeds and total customer cart value:

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

Customers choosing express delivery show a **3.46% higher willingness to spend** per order compared to those using standard options. This confirms that speed-sensitive shoppers are a premium segment. Upselling delivery options during checkout provides a direct path to increasing average order value.

5. AGE GROUP REVENUE CONTRIBUTION

Binning consumer ages reveals how revenue is distributed across different lifecycles, showing stable and consistent purchasing power across generations.

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

Strategic Takeaway

The data shows a uniform revenue distribution across age groups, with less than a 10.3% variance between the highest-earning segment (Young Adults) and the lowest (Seniors). Rather than tailoring product lines to specific generations, marketing campaigns should focus on cross-generational themes like lifestyle preferences and shopping frequency.

6. TARGETED BUSINESS DIRECTIVES

Revitalize Subscription Perks

The core subscription offering needs to pivot away from flat transactional discounts. Because member average spend matches non-member spend almost exactly, the current program is not encouraging larger cart sizes. Tiered retention models should introduce exclusive value-add benefits—such as early access to product drops or free upgrades to Express Shipping—to improve margins while driving purchase frequency.

Deploy an Automation Funnel for "Returning" Customers

The dataset shows an immediate growth opportunity within customer segmentation.

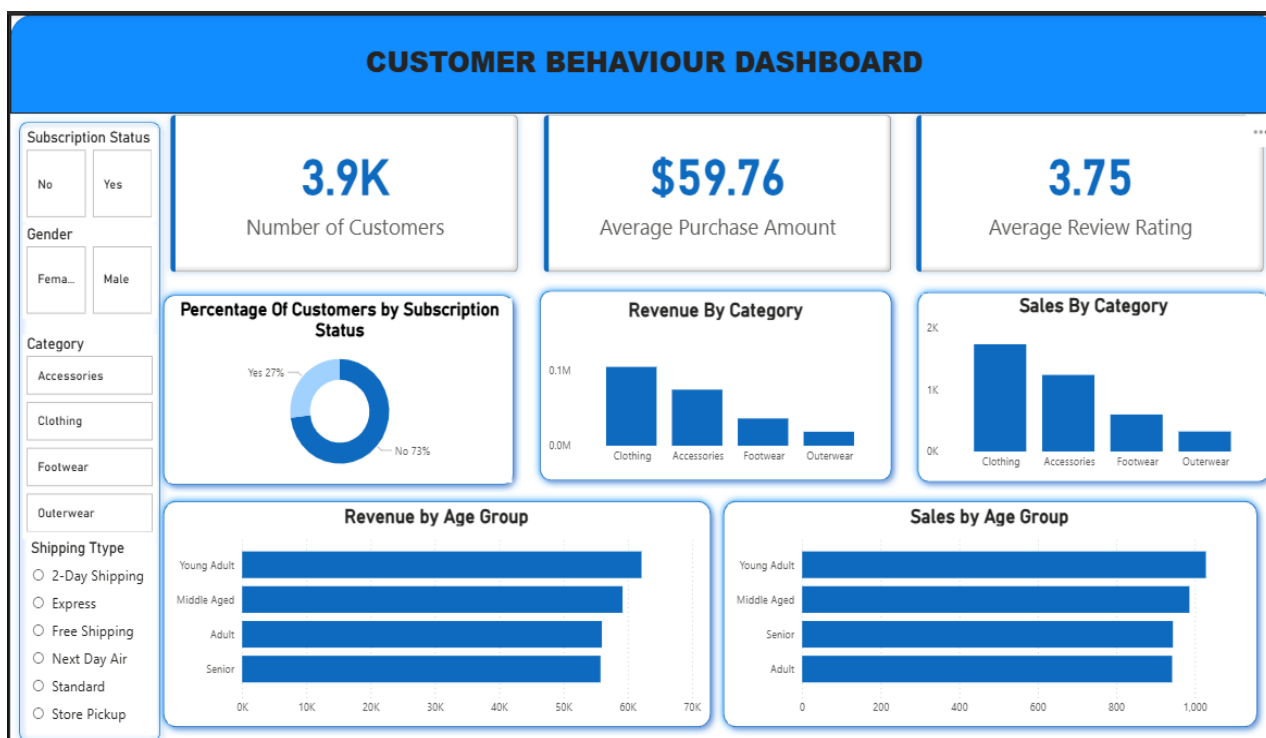
	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

With a solid base of 3,116 loyal users, the platform has a stable revenue core. Strategic focus should now shift to the **701 returning customers**. Moving just 15% of this mid-tier group into the loyal category via personalized lifecycle marketing would expand the platform's high-value user base by more than 100 customers.

Optimize Promotional Margins

Anchor products like hats and sneakers are over-exposed to discounts, with nearly half of their sales volumes relying on promotional codes. The platform should reduce generic, site-wide discounts and replace them with data-driven bundling strategies (e.g., pairing high-margin accessories with top-volume clothing items) to protect product profitability.

6. DASHBOARD IN POWER BI



7. BUSINESS RECOMMENDATIONS

- **Boost Subscriptions** – Promote exclusive benefits for subscribers.
- **Customer Loyalty Programs** – Reward repeat buyers to move them into the “Loyal segment”.
- **Review Discount Policy** – Balance sales boosts with margin control.
- **Product Positioning** – Highlight top-rated and best-selling products in campaigns.
- **Targeted Marketing** – Focus efforts on high-revenue age groups and express-shipping users.